LAB EXPERIMENT RECORD

Set 14: Survey Results

Department	AI&DS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using Tableau
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Day	Thursday

Aim

To visualize survey results using a stacked bar chart, a radar chart and a table, and to build an interactive Tableau dashboard for exploring the survey responses.

Dataset: Survey Results

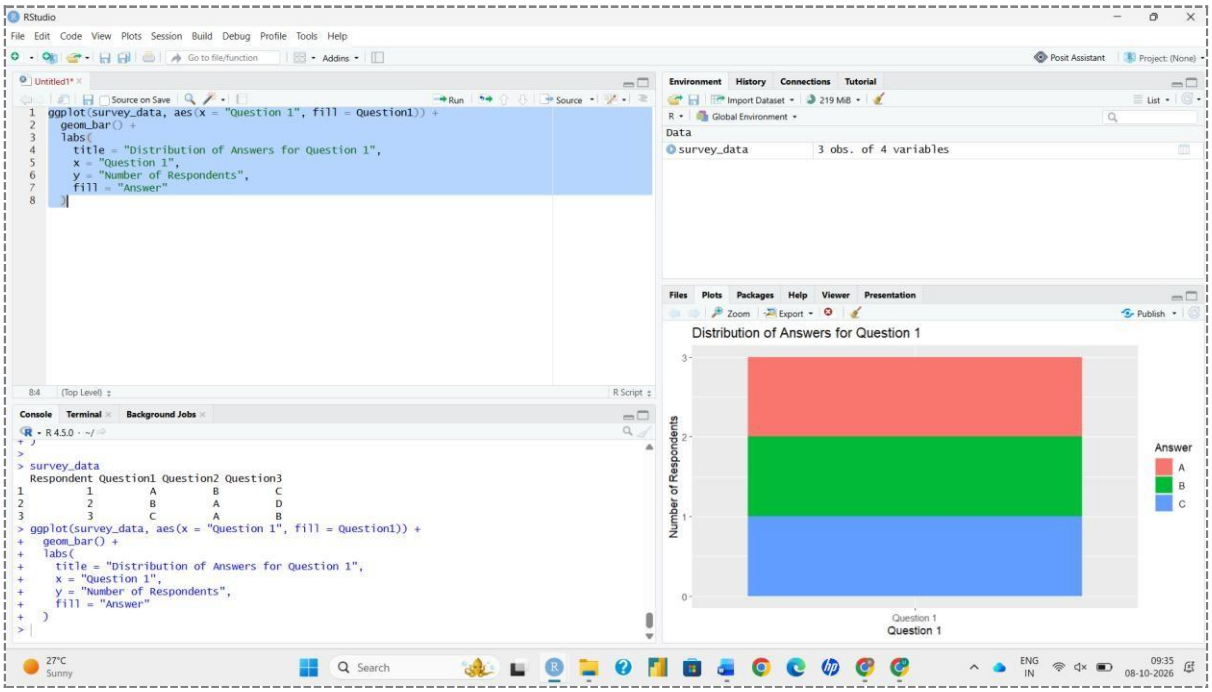
Respondent	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

PROGRAM 1: Stacked Bar Chart for Question 1

Create a stacked bar chart to display the distribution of answers for Question 1. Label the chart elements.

Steps in Tableau:

1. Connect to the survey data file and open a new sheet named "Q1 Stacked Bar".
2. Drag Question 1 to Color (this splits the bar by answer A, B and C).
3. Drag Respondent to Rows and change it to Count (CNT) so the bar height shows the number of respondents.
4. Drag Question 1 to Label (or Count to Label) to show the value inside each segment.
5. Set the title to "Distribution of Answers for Question 1", label the Y-axis "Number of Respondents" and show the colour legend.



Output 1: Stacked Bar Chart of Question 1 Responses

Observation: Each answer (A, B and C) was chosen by exactly one respondent, so the stacked bar is divided into three equal segments and no option dominates Question 1.

PROGRAM 2: Radar Chart of Overall Response Pattern

Generate a radar chart to visualize the overall pattern of responses across all three questions.

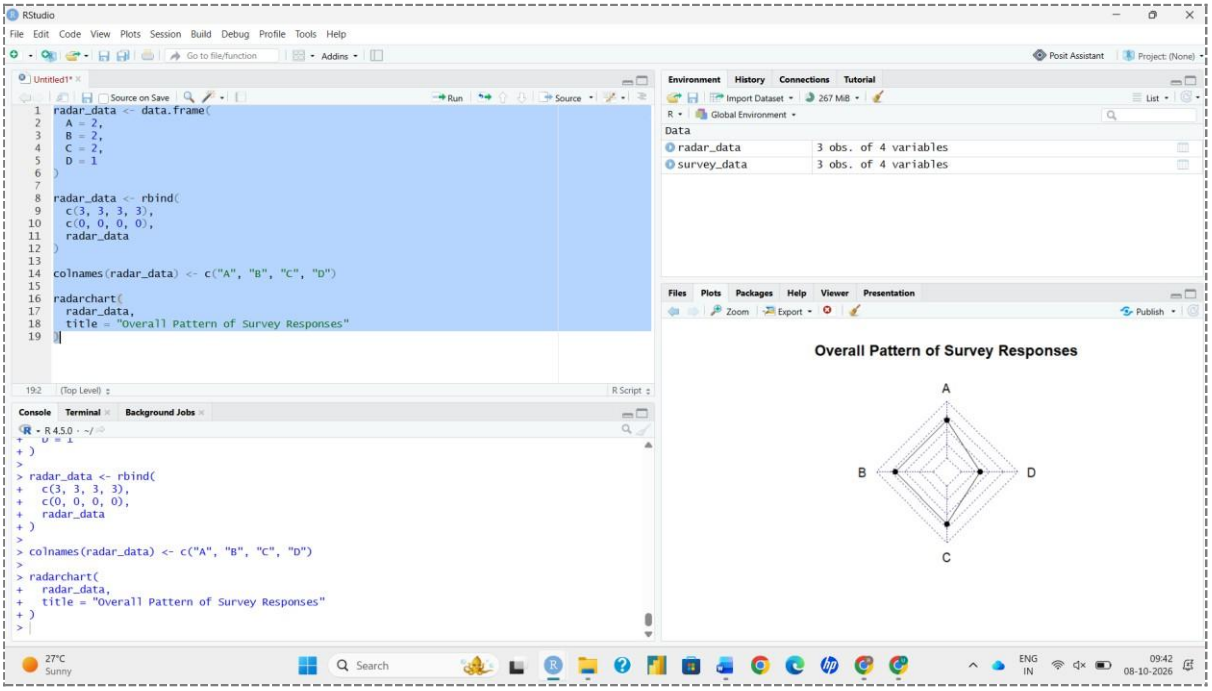
Steps in Tableau:

1. Tableau has no built-in radar chart, so it is built from a path/polygon mark using calculated fields.
2. Prepare the response count summary below (answers counted per question) as a second sheet or by pivoting Question 1-3 into Question / Answer columns.
3. Create calculated field Axis Index: CASE [Question] WHEN "Question 1" THEN 0 WHEN "Question 2" THEN 1 ELSE 2 END.
4. Create Angle = [Axis Index] * 2 * PI() / 3, then X = [Count] * COS([Angle]) and Y = [Count] * SIN([Angle]).
5. Drag X to Columns and Y to Rows, set the mark type to Line (or Polygon), put Option on Color and Axis Index on Path.
6. Repeat the first axis point at the end of each series to close the shape, add axis labels (Question 1-3) and the title "Overall Response Pattern".

Response count summary (used for the radar chart):

Answer Option	Question 1	Question 2	Question 3
A	1	2	0
B	1	1	1
C	1	0	1

Answer Option	Question 1	Question 2	Question 3
D	0	0	1



Output 2: Radar Chart of Survey Response Pattern

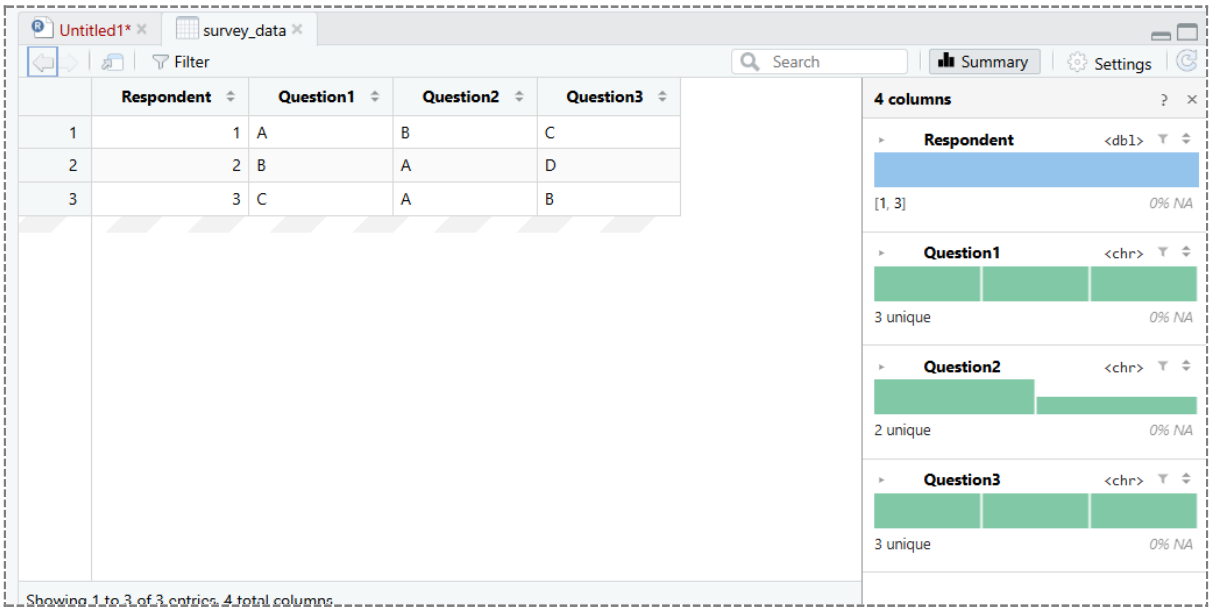
Observation: Option A is strongest on Question 2 (chosen by 2 of 3 respondents), while option B is the only answer chosen once on every question. Question 3 shows the most varied answers (B, C and D each once), so responses are spread out rather than concentrated.

PROGRAM 3: Table of Survey Responses

Build a table to show the survey response data for each respondent. Label the table elements.

Steps in Tableau:

1. Create a new sheet named "Survey Table".
2. Drag Respondent to Rows.
3. Pivot Question 1, Question 2 and Question 3 (or use Measure Names) and drag the question names to Columns.
4. Drag the answer values to Text so each cell shows the respondent's answer.
5. Add the title "Survey Responses by Respondent" and keep row and column headers visible.



Output 3: Table of Survey Responses

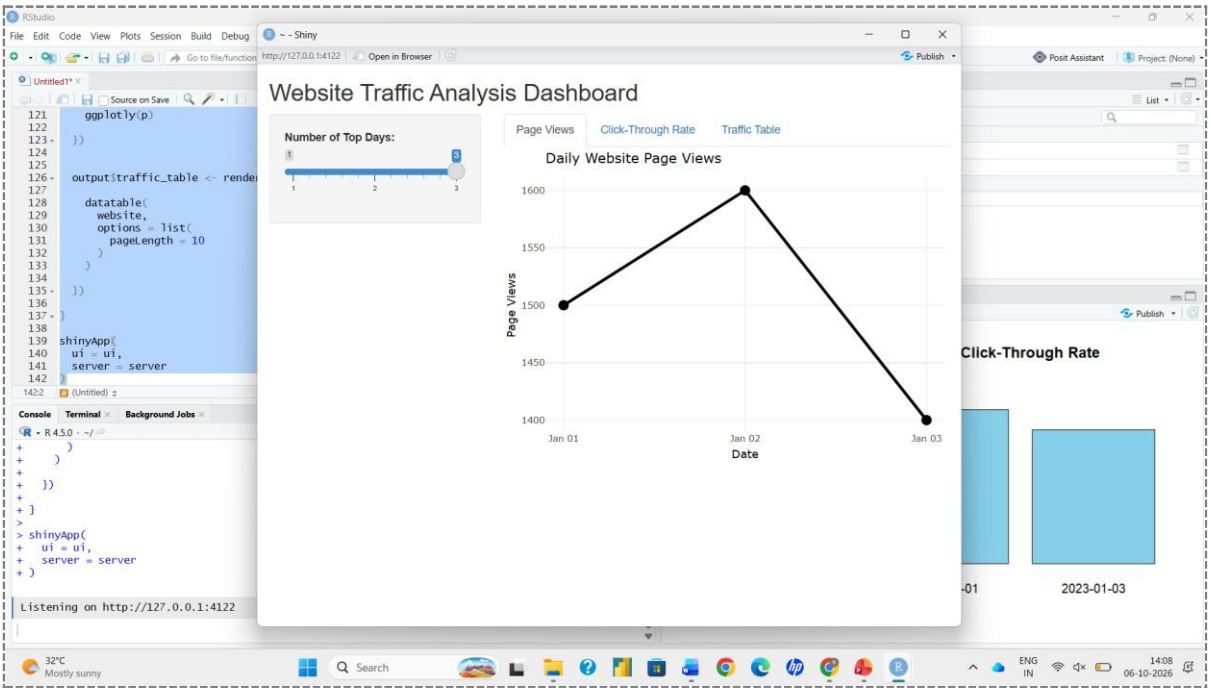
Observation: The table shows the exact answer given by each respondent to each question, which complements the summarised charts.

PROGRAM 4: Interactive Tableau Dashboard

Develop a Tableau dashboard combining the stacked bar chart, radar chart and table for interactive exploration of survey responses.

Steps in Tableau:

1. Click New Dashboard at the bottom of the workbook and rename it.
2. Set Size to Automatic (or Fixed: Desktop Browser) from the Dashboard pane.
3. Drag Q1 Stacked Bar, Radar Chart and Survey Table from the Sheets list onto the dashboard canvas.
4. Add a Title (Dashboard > Show Title) and arrange the sheets using Tiled layout containers.
5. Select the first chart and click Use as Filter so that selecting an answer segment filters the radar chart and the table.
6. Use Dashboard > Actions to add a Highlight action, and enable Show Filters on the sheets for dropdown filtering.
7. Click different marks to test interactivity and confirm all views update together.



Output 4: Interactive Survey Responses Dashboard

Observation: Clicking an answer in the stacked bar highlights the matching respondents in the table and updates the radar chart, making it easy to explore individual and overall responses.

Procedure

1. Load the survey data into Tableau.
2. Build the stacked bar chart for Question 1 using Question 1 on Color and count of Respondent.
3. Build the radar chart from the response counts using calculated fields for X and Y coordinates.
4. Build the table showing each respondent's answers.
5. Combine the three sheets in a dashboard and enable filter and highlight actions.
6. Test the dashboard by selecting different answers.

Result

The survey results were successfully visualized with a stacked bar chart, radar chart and table, and an interactive Tableau dashboard was created for exploring the responses.